American Computer Science League

2025-2026 ● Contest 1: Creature Capture ● Senior Division

PROBLEM: A creature is hiding in a cell of a 256 x 256 grid. You will be given the location of 3 guesses and how far away the creature is from each guess. Finding these possible locations for each guess will produce a common location where the creature is hiding. Find that location.

Each cell in the grid is represented by a 4-digit hexadecimal string. The first 2 digits signify the row (00-FF, from top to bottom), and the last 2 digits signify the column (00-FF, from left to right).

How far the guess is from the creature's location is called the *distance value*. It is a 4-digit hexadecimal string, where the first 2 digits and the last 2 digits indicate how far away the creature is from the guess horizontally or vertically. For example, a distance value of 3A05 means that the creature could be 3A (58_{10}) rows away and 5 columns away, or 5 rows away and 3A (58_{10}) columns away. There are 8 possible locations, though some of those locations might be off the grid and thus, ignored.

After finding all possible cells in the grid where the creature could be hiding for each guess, output the location where the creature is hiding in (row,column) format.

EXAMPLE:

Input	`4D93 414D`
	`A566 997A`
	`F633 EAAD`

Output	`(0C,E0)`
Explanation	The user's first guess is location row 4D and column 93. The distance value is 41 rows away and 4D columns away or 4D rows away and 41 columns away from the creature. The other 2 guesses are done the same way. ● For guess (4D,93), the possible locations are (8E,46), (8E,E0), (0C,46), (0C,E0), (9A,52), (9A,D4), (00,52), (00,D4). ● For guess (A5,66), the possible locations are (0C,E0), (2B,FF). ● For guess (F6,33), the only possible location is (0C,E0). The only location that is common for all 3 guesses is (0C,E0). Output `(0C,E0)`.

INPUT: The input will consist of 3 strings representing 3 guesses. Each string consists of two 4-digit hexadecimal numbers, where the first number is the row and column location of the guess and the second is the distance value.

OUTPUT: Output the ordered pair in (row,column) format representing where the creature is hiding.

SAMPLE INPUT:

```
1. 4D93 414D
   A566 997A
   F633 EAAD
2. 093C A396
   D675 2A5D
   397E 7354
3. 4C0D 3334
   33BB 4C7A
   EC8E 6D4D
4. 16FA AC24
   57FD 6B27
   63B4 5F22
5. B531 7760
   C8FA 8A69
   F543 B74E
6. EF65 207B
   6F4A 6096
   43C8 8C18
```

SAMPLE OUTPUT:

```
1. (0C,E0)
2. (AC,D2)
3. (7F,41)
4. (C2,D6)
5. (3E,91)
6. (CF,E0)
```

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TEST DATA

TEST INPUT:

```
1. FB2B 055A
   348E C209
   C50D 3178
2. C293 4C25
   F3F5 7D87
   DA02 646C
3. 809D 6813
   2A9A 1210
   EED9 D64F
4. DEA5 A601
   37F3 014D
   5935 2171
5. BCD5 3A4F
   3E0B B87B
   3975 BD11
6. DB8B 0D26
   BDDE 112D
   22FF AC4E
```

TEST OUTPUT:

```
1. (F6,85)
2. (76,6E)
3. (18,8A)
4. (38,A6)
5. (F6,86)
6. (CE,B1)
```

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问题：一个生物藏在一个 256×256 网格**的某个单元格中。现给出** 3 **个猜测位置（生物可能所在单元格），以及该生物分别离这三个**猜测**位置的距离。**请推算出每次猜测对应的所有可能藏匿位置，再从中找出所有猜测结果共有的位置。该位置即为目标物的实际藏匿点，请找出此位置。

网格中的每个单元格由一个 4 **位十六进制字符串表示，前两位数字代表行（**00-FF**，从上到下），后两位数字代表列（**00-FF**，从左到右）。**

猜测位置与生物实际藏匿**位置之间的距离称为*距离值*。它也是一个** 4 **位十六进制字符串，其中前两位和后两位分别表示生物在水平或垂直方向上**与**猜测位置的**距离**。例如，距离值** 3A05 意味着**该生物可能在** 3A (58_{10}) 行、5 列的**位置，或者在** 5 行、 3A (58_{10})列的位置。总共有 8 个可能的坐标位置，不过其中一些位置可能超出网格范围，因此会被忽略。

在找出每个猜测位置对应的所有可能单元格后，请以(行,列)的格式输出该生物的实际藏匿**位置。**

示例：

输入	`4D93 414D`
	`A566 997A`
	`F633 EAAD`

输出	`(0C,E0)`
解析	该**用户的第一个猜测位置为第** 4D 行**、第** 93 **列。**距离值表明，该生物的实际藏匿位置可能与该猜测位置相距 41 行 4D 列，或 4D 行 41 列。**其余两个猜测位置**也按相同方式处理： ● 猜测**位置** (4D,93)，**可能的生物位置有** (8E,46), (8E,E0),(0C,46), (0C,E0), (9A,52), (9A,D4), (00,52), (00,D4)。 ● 猜测位置(A5,66), **可能的生物位置有** (0C,E0), (2B,FF)。 ● 猜测位置(F6,33), **唯一可能的生物位置是** (0C,E0)。 三个**猜测位置的唯一共同生物位置是**(0C,E0)。**输出** `(0C,E0)`.

输入：输入将包含 3 **个字符串，代表** 3 **个猜测位置。每个字符串由两个** 4 **位十六进制数字组成，其中第一个数字表示猜测的行和列位置，第二个数字表示距离值。**

输出：输出表示生物实际隐藏位置的有序对，格式为 (行,列)。

样例输入：

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数据测试

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